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Inventor(s)	CHAN; Albert P. et al.

CELL SENSING STRUCTURE AND METHODS OF FORMATION

Abstract

Implementations described herein relate to various structures, integrated assemblies, and memory devices. In some implementations, a structure includes a semiconductor layer, a dielectric layer that is proximate to the semiconductor layer, and a multi-layer structure that extends away from an approximately planar surface that is across the semiconductor layer and the dielectric layer. The multi-layer structure includes a planarized tip region and an outer silicon oxycarbide layer that extends from the approximately planar surface to the planarized tip region. In some implementations, a portion of the outer silicon oxycarbide layer in the planarized tip region is unadulterated. The structure further includes a conductive structure proximate to the multi-layer structure. The conductive structure includes a portion that extends through the approximately planar surface and has a tip with an isotropic morphology. In some implementations, the isotropic morphology includes a curvature traversing a portion of the semiconductor layer and a portion of the dielectric layer.

Inventors: **CHAN; Albert P. (Boise, ID), RUI; Ying (Meridian, ID), CHEN; Meng-Hsien (Taichung City, TW), LIN; Yenting (Taichung City, TW), BARR; Elisabeth (Boise, ID), WU; Chongzan (Taichung City, TW)**

Applicant: **Micron Technology, Inc.** (Boise, ID)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This patent application claims priority to U.S. Provisional Patent Application No. 63/559,825, filed on Feb. 29, 2024, entitled "CELL SENSING STRUCTURE AND METHODS OF FORMATION," and assigned to the assignee hereof. The disclosure of the prior application is considered part of and is incorporated by reference into this patent application.

TECHNICAL FIELD
[0002] The present disclosure generally relates to semiconductor devices and methods of forming semiconductor devices. For example, the present disclosure relates to a cell sensing structure for a memory device.
BACKGROUND
[0003] Memory devices are widely used to store information in various electronic devices. A memory device includes memory cells. A memory cell is an electronic circuit capable of being programmed to a data state of two or more data states. For example, a memory cell may be programmed to a data state that represents a single binary value, often denoted by a binary "1" or a binary "0." As another example, a memory cell may be programmed to a data state that represents a fractional value (e.g., 0.5, 1.5, or the like). To store information, the electronic device may write, or program, a set of memory cells. To access the stored information, the electronic device may read, or sense, the stored state from the set of memory cells.
[0004] Various types of memory devices exist, including random access memory (RAM), read only memory (ROM), dynamic RAM (DRAM), static RAM (SRAM), synchronous dynamic RAM (SDRAM), ferroelectric RAM (FeRAM), magnetic RAM (MRAM), resistive RAM (RRAM), flash memory (e.g., NAND memory and NOR memory), and others. A memory device may be volatile or non-volatile. Non-volatile memory (e.g., flash memory) can store data for extended periods of time even in the absence of an external power source. Volatile memory (e.g., DRAM) may lose stored data over time unless the volatile memory is refreshed by a power source. A binary memory device may, for example, include a charged or discharged capacitor. A charged capacitor may, however, become discharged over time through leakage currents, resulting in the loss of the stored information. Some features of volatile memory may offer advantages, such as faster read or write speeds, while some features of non-volatile memory, such as the ability to store data without periodic refreshing, may be advantageous.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] FIG. **1** is a circuit diagram of an example memory cell described herein.
[0006] FIG. **2** is a diagrammatic view of a cell sensing structure described herein.
[0007] FIG. **3** is a flowchart of an example method of forming an integrated assembly or memory device having the cell sensing structure.
[0008] FIGS. **4**A through **4**H are diagrammatic views showing formation of the structure at example process stages of an example process of forming the cell sensing structure.

[0009] FIG. **5** is a diagrammatic view of example etch operations described herein.
[0010] FIG. **6** is a diagrammatic view of an example memory device described herein.

DETAILED DESCRIPTION

[0011] In a semiconductor device, a DRAM memory cell may be accessed using a combination of conductive structures that include a digit line structure and a cell contact structure. The digit line structure and the cell contact structure may be above an underlying active area of the semiconductor device (e.g., an area of the semiconductor device including active circuitry such as a transistor including source and/or drain regions). Techniques to fabricate the digit line structure and the cell contact structure may include using a series of semiconductor manufacturing operations to sequentially deposit, pattern, and/or etch a combination of conductive and/or dielectric layers over the underlying active area.

[0012] In some cases, the digit line structure is formed prior to the cell contact structure. Forming the digit line structure may include forming a multi-layer structure that includes one or more dielectric layers that electrically isolate a conductive layer (e.g., a digit line) of the digit line structure from another conductive layer of an adjacent digit line structure. Forming the multi-layer structure may further include forming a cap portion over a tip of the digit line structure and a footer portion that protrudes laterally from a base portion of the digit line structure directly over the underlying active area.

[0013] After the digit line structure is formed, a dry etch operation may be used to "punch" a cavity through the footer portion and into the active area for formation of the cell contact structure. However, while forming the cavity, the dry etch operation may thin and/or remove the cap. The thinning and/or removal of the cap may reduce process windows of subsequent semiconductor processing operations (e.g., subsequent etch operations used to form additional features of the semiconductor device), causing the digit line structure to be susceptible to additional material removal that eventually exposes the conductive layer to cause a potential of electrical shorting and reduce a quality and/or reliability of the semiconductor device.

[0014] Some implementations described herein include a semiconductor device that includes a cell sensing structure having a digit line structure and an adjacent cell contact structure. Techniques to form the digit line structure include forming a multi-layer structure including a cap portion and a base portion, where a footer portion extends laterally from the base portion. The techniques include forming a temporary, non-conformal layer over the cap portion during removal of the footer portion. After removal of the footer portion, the techniques include using a wet-etch operation to punch a cavity for the cell contact structure into an underlying active area.

[0015] In this way, the cap portion is preserved to increase a process margin associated with subsequent formation of one or more additional features of the semiconductor device, thereby reducing a likelihood of the conductive layer being exposed through removal of materials included in one or more layers of the digit line structure. Furthermore, and by reducing the likelihood of the conductive layer being exposed, a potential of electrical shorting within the semiconductor device is reduced, to improve a quality and/or a reliability of the semiconductor device, thereby reducing an amount of resources (e.g., raw materials, labor, semiconductor processing tools, and/or computing resources) used to support a market consuming the semiconductor device.

[0016] FIG. **1** is a circuit diagram of an example memory cell **100** described herein. In some implementations, the memory cell **100** is a ferroelectric memory cell. Alternatively, the memory cell **100** may be a linear dielectric memory cell or a paraelectric memory cell. As shown in FIG. **1**, the memory cell **100** may include a transistor **105** (or another type of selection circuit) and a capacitor **110**. The memory cell **100** may be accessed (e.g., written to, read from, and/or erased) using signals on a combination of lines that are coupled to the memory cell **100**, shown as an access line **115** (sometimes called a "word line"), a digit line **120** (sometimes called a "bit line"), and a plate line **125**.

[0017] The transistor **105** (sometimes called an access transistor) may include a gate **130**. The capacitor **110** includes a bottom electrode **135** and a top electrode **140** separated by an insulator **145**. In some implementations, the capacitor is a ferroelectric capacitor, and the insulator **145** is a ferroelectric insulator that comprises, consists of, or consists essentially of ferroelectric material. Alternatively, the capacitor may be a linear dielectric capacitor, and the insulator **145** may be a linear dielectric insulator that comprises, consists of, or consists essentially of linear dielectric material. Alternatively, the capacitor may be a paraelectric capacitor, and the insulator **145** may be a paraelectric insulator that comprises, consists of, or consists essentially of paraelectric material. When the access line **115** is activated (e.g., when a voltage is applied to the access line **115**), the gate **130** coupled to the access line **115** may be activated. When the gate **130** is activated, the transistor **105** couples the digit line **120** to the bottom electrode **135** of the capacitor **110**. A state of the memory cell **100** may then be written or read via the digit line **120**.

[0018] The top electrode **140** of the capacitor **110** may be coupled to the plate line **125** and a cell plate **150**. To write to (or program) the memory cell **100**, the access line **115** may be activated, and a voltage may be applied across the capacitor **110** by controlling the voltage of the top electrode **140** (via the plate line **125** and/or the cell plate **150**) and/or the bottom electrode **135** (via the digit line **120**).

[0019] In some implementations, data may be stored using the capacitor **110** by controlling a voltage difference and/or a polarity difference of the capacitor **110** (e.g., of the insulator **145** between the bottom electrode **135** and the top electrode **140**). For example, a voltage of the cell plate **150** and the digit line **120** may be controlled. In some implementations, a negative polarity of the insulator **145** as compared to the cell plate **150** results in a logic “0” state being stored in the capacitor **110**, and a positive polarity of the insulator **145** as compared to the cell plate **150** results in a logic “1” state being stored in the capacitor **110**. For a linear dielectric capacitor or a paraelectric capacitor, the cell plate **150** may grounded, and the capacitor **110** may be charged by applying a voltage to the bottom electrode **135** via the digit line **120**.

[0020] To read the memory cell **100** (e.g., a state stored by the capacitor **110**), the access line **115** may be activated, and a voltage may be applied to the plate line **125**. Applying a voltage to the plate line **125** may cause a change in the stored charge on the capacitor **110**. The magnitude of the change in stored charge may depend on the stored state of capacitor **110** (e.g., whether the stored state is a logic “1” state or a logic “0” state). This may or may not induce a threshold change in the voltage of the digit line **120** based on the charge stored on the capacitor **110**. The change in voltage or lack of change in voltage of the digit line **120** (or a magnitude of the change in voltage) may be used to determine the stored state of the capacitor **110**. For example, if the change in voltage satisfies a threshold, then the read operation indicates that a first state was stored in the capacitor **110**, whereas if the change in voltage does not satisfy the threshold, then the read operation determines that a second state was stored in the capacitor **110**. In some cases, multiple threshold voltages may be used, such as when the capacitor is capable of storing more than two data states (e.g., for a multi-level cell, a triple-level cell, and so on).

[0021] In some implementations, the memory cell **100** is accessed using a cell contact **155** that is part of a connection between the transistor **105** and the capacitor **110**. As described in greater detail in connection with FIGS. **2-6**, a cell sensing structure may include the digit line **120** and the cell contact **155**. The cell sensing structure may be a multi-layer structure having a combination of dielectric layers and conductive layers and include a first structure that includes the digit line **120** and a second, adjacent structure that includes the cell contact **155**. The first structure (e.g., a digit line structure) may include a tip region having a cap portion that covers, protects, and/or electrically isolates the digit line **120**. During fabrication of the cell contact **155**, preservation of the cap portion is crucial to prevent electrical shorting of the digit line **120** with other conductive structures and/or integrated circuitry in a semiconductor device including the memory cell **100**.

[0022] As indicated above, FIG. **1** is provided as an example. Other examples may differ from what is described with respect to FIG. **1**.

[0023] FIG. **2** is a diagrammatic view of a cell sensing structure **200** described herein. The cell sensing structure **200** may be part of an integrated assembly, such as a memory array, a portion of a memory array, or a dynamic random access memory device that includes the memory array and one or more other components (e.g., sense amplifiers, a row decoder, a column decoder, a row address buffer, a column address buffer, one or more data buffers, one or more clocks, one or more counters, and/or a memory controller).

[0024] The cell sensing structure **200** (e.g., a multi-layer cell sensing structure) may include combinations of one or more semiconductive materials (e.g., one or more semiconductor layers), dielectric materials (e.g., one or more dielectric layers), and/or conductive materials (e.g., one or more conductive layers). Semiconductive materials may comprise, consist of, or consist essentially of silicon (e.g., polycrystalline silicon), silicon germanium, gallium arsenide, indium phosphide, gallium nitride, silicon carbide, or a type III-V element, among other examples. Dielectric materials may comprise, consist of, or consist essentially of silicon dioxide, silicon nitride, silicon oxycarbide, and/or aluminum dioxide, among other examples. Conductive materials may comprise, consist of, or consist essentially of a metal (e.g., titanium, tungsten, cobalt, nickel, platinum, and/or ruthenium), a metal composition (e.g., a metal silicide, a metal carbide, and/or a metal nitride, such as titanium nitride or titanium silicon nitride), and/or a conductively-doped semiconductive material.

[0025] The cell sensing structure **200** includes a contact structure **205** (e.g., a conductive structure corresponding to the cell contact **155** of FIG. **1**) between multi-layer structures **210-1** and **210-2** (e.g., between digit line structures). The contact structure **205** includes combinations and/or portions of conductive layers. For example, and as shown in FIG. **2**, the contact structure **205** includes a conductive layer **215-1** that is on and/or under a conductive layer **215-2**. Furthermore, the contact structure **205** electrically couples with a capacitor (e.g., the capacitor **110** of FIG. **1**).

[0026] Each of the multi-layer structures **210-1** and **210-2** may include combinations and/or portions of conductive layers, semiconductive layers, and/or dielectric layers. Using the multi-layer structure **210-1** as an example, and as shown in FIG. **2**, the multi-layer structure **210** includes the conductive layer **215-3** (e.g., corresponding to the digit line **120** of FIG. **1**) that is on and/or over the conductive layer **215-4** (e.g., corresponding to a portion of the gate **130** of FIG. **1**).

[0027] The multi-layer structure **210-1** may further include dielectric layers **220-1**, **220-2**, **220-3**, **220-4**, **220-5**, **220-6**, and **220-7**. One or more of the dielectric layers **220-1** through **220-7** may be over and/or on the conductive layers **215-3** and **215-4**, and electrically insulate the conductive layers **215-3** and **215-4** (e.g., prevent electrical shorting between the conductive layers **215-3** and **215-4** and integrated circuitry, or other conductive layers, of a semiconductor device including the cell sensing structure **200**).

[0028] In some implementations, the dielectric layer **220-7** is an outer layer that includes a silicon oxycarbide material (e.g., the dielectric layer **220-7** is an outer silicon oxycarbide layer). As described in greater detail in connection with FIGS. **4**A through **4**H and elsewhere herein, techniques to form the cell sensing structure **200** may include forming a cap portion that includes the dielectric layer **220-7** (e.g., the silicon oxycarbide material). The techniques may further include forming a temporary, non-conformal layer over the cap portion to preserve the dielectric layer **220-7** during an etching operation that forms a cavity for the contact structure **205**. Preserving the dielectric layer **220-7** during the etching operation may offer etch selectivity benefits during subsequent operations to form a device that includes the cell sensing structure **200**.

[0029] As further shown in FIG. **2**, the multi-layer structure **210-1** has a base region **225** and a planarized tip region **230**. The planarized tip region **230** (e.g., a distal tip of the multi-layer structure **210-1**) may be away from the base region **225**.

[0030] Portions of the dielectric layers **220-5**, **220-6**, and **220-7** may have different arrangements throughout the multi-layer structure **210-1**. For example, the dielectric layers **220-5**, **220-6**, and/or **220-7** may each extend from the base region **225** to the planarized tip region **230**. Further, and in some implementations, portions of the dielectric layers **220-5**, **220-6**, and/or **220-7** within the planarized tip region **230** are unadulterated (e.g., portions of the dielectric layers **220-5**, **220-6**, and/or **220-7** within the planarized tip region **230** may, within reasonable process control capabilities, be free of dopants and/or other impurities that are intentionally formed within and/or on the layers **220-5**, **220-6**, and/or **220-7** to alter and/or change one or more physical properties of the dielectric layers **220-5**, **220-6**, and/or **220-7**).

[0031] As further shown in FIG. **2**, portions of the dielectric layers **220-5**, **220-6**, and **220-7** between the base region **225** and the planarized tip region **230** may combine to form a multi-layer sidewall structure **235** (e.g., a vertical sidewall structure). In some implementations, the multi-layer sidewall structure **235** is between the conductive layer **215-3** (e.g., a digit line) and the contact structure **205** (e.g., a cell contact structure).

[0032] A device region **240** that includes one or more active devices (e.g., the transistor **105** of FIG. **1**) may be proximate (e.g., under) the base region **225**. As shown in FIG. **2**, the device region **240** includes a dielectric layer **220-8**, a dielectric layer **220-9**, and a semiconductor layer **245**. In some implementations, the dielectric layer **220-9** and the semiconductor layer **245** share an approximately planar surface (e.g., a top surface) that extends across the base region **225**.

[0033] In some implementations, the semiconductor layer **245** is an active region (e.g., an active area) that includes one or more portions of a transistor device (e.g., a source/drain region of the transistor **105**). Additionally, or alternatively and in some implementations, the dielectric layer **220-9** is an insulative region (e.g., insulative area) between one or more active areas of the device region **240**.

[0034] As shown in FIG. **2**, the contact structure **205** (e.g., the conductive layer **215-2**) extends into the semiconductor layer **245** (e.g., the cell contact **155** makes an electrical connection with the transistor **105**). In some implementations, the contact structure **205** extends into the dielectric layer **220-9** (e.g., extends into the dielectric layer **220-9** that is proximate and/or laterally adjacent to the semiconductor layer **245**). Additionally, or alternatively and in some implementations, the contact structure **205** is approximately parallel to the dielectric layer **220-7**.

[0035] As further shown in FIG. **2**, and as described in greater detail in connection with FIG. **5**, a tip of the contact structure **205** includes an isotropic morphology **250**. The isotropic morphology **250** includes a curvature that traverses portions of the dielectric layer **220-9** and the semiconductor layer **245**. In some implementations, an inflection point at which the contact structure **205** transitions to the isotropic morphology (e.g., an inflection point near the tip) aligns with an outer surface of the dielectric layer **220-7**.

[0036] The multi-layer structure **210-2** may be similar to the multi-layer structure **210-1**. For example, the multi-layer structure **210-2** may include one or more of the dielectric layers **220-1** through **220-7** and the conductive layer **215-3**. Additionally, or alternatively, the multi-layer structure **210-2** may include additional dielectric layers and/or exclude the conductive layer **215-4**.

[0037] As indicated above, FIG. **2** is provided as an example. Other examples may differ from what is described with regard to FIG. **2**.

[0038] As described in connection with FIGS. **1** and **2**, and in some implementations, a structure (e.g., the cell sensing structure **200**) includes a semiconductor layer (e.g., the semiconductor layer **245**), a dielectric layer (e.g., the dielectric layer **220-9**) that is proximate (e.g., laterally adjacent) to the semiconductor layer, and a multi-layer structure (e.g., the multi-layer structure **210-1**) that extends away from an approximately planar surface (e.g., the base region **225**) that is across the semiconductor layer and the dielectric layer. The multi-layer structure includes a planarized tip region (e.g., the planarized tip region **230**) and an outer silicon oxycarbide layer (e.g., the dielectric layer **220-7**) that extends from the approximately planar surface to the planarized tip region. In some implementations, a portion of the outer silicon oxycarbide layer in the planarized tip region is unadulterated. The structure further includes a conductive structure (e.g., the contact structure **205**) proximate (e.g., laterally adjacent) to the multi-layer structure. The conductive structure includes a portion that extends through the approximately planar surface and has a tip with an isotropic morphology (e.g., the isotropic morphology **250**). In some implementations, the isotropic morphology includes a curvature traversing a portion of the semiconductor layer and a portion of the dielectric layer.

[0039] In some implementations, an apparatus (e.g., a memory device including the cell sensing structure **200**) includes a device region (e.g., the device region **240**) and a digit line structure that extends away from the device region. The digit line structure includes a base region (e.g., the base region **225**), a planarized tip region (e.g., the planarized tip region **230**) that is away from the base region, and a multi-layer structure (e.g., the multi-layer structure **210-1**) including a first dielectric layer (e.g., the dielectric layer **220-5**) that extends from the base region to the planarized tip region, a second dielectric layer (e.g., the dielectric layer **220-6**) that conforms to the first dielectric layer and that extends from the base region to the planarized tip region, and a third dielectric layer (e.g., the dielectric layer **220-7**) that conforms to the second dielectric layer and that extends to form the base region to the planarized tip region. In some implementations, portions of the first dielectric layer, the second dielectric layer, and the third dielectric layer that are in the planarized tip region are unadulterated. The apparatus further includes a cell contact structure (e.g., the contact structure **205**) that is proximate (e.g., laterally adjacent) to the digit line structure. The cell contact structure includes a tip that has an isotropic morphology (e.g., the isotropic morphology **250**) that extends beyond the base region and into an insulative area (e.g., the dielectric layer **220-9**) and an active area (e.g., the semiconductor layer **245**) within the device region. In some implementations, the isotropic morphology includes a curvature traversing a portion of the insulative area and a portion of the active area.

[0040] FIG. **3** is a flowchart of an example method **300** of forming an integrated assembly or memory device having a cell sensing structure described herein (e.g., the cell sensing structure **200**). In some implementations, and as described in greater detail in connection with FIGS. **4**A through **4**H, one or more process blocks of FIG. **3** may be performed by various semiconductor manufacturing equipment.

[0041] As shown in FIG. **3**, the method **300** may include forming, above a device region (e.g., the device region **240**) of a memory device (e.g., the memory cell **100**), a digit line structure including a multi-layer structure (e.g., the multi-layer structure **210-1**) having a footer portion extending away from a base region (e.g., the base region **225**) of the multi-layer structure and a cap portion in a tip region (e.g., the planarized tip region **230**) that extends away from the base region (block **310**). As further shown in FIG. **3**, the method **300** may include forming a temporary, non-conformal layer over the cap portion (block **320**). As further shown in FIG. **3**, the method **300** may include damaging a top segment of the footer portion (block **330**). As further shown in FIG. **3**, the method **300** may include removing the temporary, non-conformal layer (block **340**). As further shown in FIG. **3**, the method **300** may include removing the footer portion (block **350**). As further shown in FIG. **3**, the method **300** may include forming a cavity that extends into an active area (e.g., the semiconductor layer **245**) of the device region (block **360**).

[0042] The method **300** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other methods described elsewhere herein.

[0043] In a first aspect, forming the multi-layer structure includes forming an outermost layer (e.g., the dielectric layer **220-7**) of the multi-layer structure using a deposition operation to deposit a layer of silicon oxycarbide over one or more dielectric layers included in the footer portion and the cap portion, wherein the silicon oxycarbide is unadulterated.

[0044] In a second aspect, alone or in combination with the first aspect, forming the temporary, non-conformal layer includes forming the temporary, non-conformal layer using a deposition operation to deposit an oxide material over the cap portion.

[0045] In a third aspect, alone or in combination with one or more of the first and second aspects, forming the temporary, non-conformal layer includes forming the temporary, non-conformal layer to have a thickness that is included in a range of approximately 4 nanometers to approximately 5 nanometers above an apex of the cap portion.

[0046] In a fourth aspect, alone or in combination with one or more of the first through third aspects, damaging the top segment of the footer portion includes damaging the top segment of the footer portion using an oxygen-based, directional bias operation.

[0047] In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, removing the temporary, non-conformal layer includes removing the temporary, non-conformal layer using a hydrofluoric acid solution.

[0048] In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, removing the footer portion includes removing a top segment of the footer portion using a hydrofluoric acid solution.

[0049] In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, removing the footer portion includes removing a bottom segment of the footer portion using a hot phosphorous solution.

[0050] In an eighth aspect, alone or in combination with one or more of the first through seventh aspects, removing the footer portion includes removing a bottom segment of the footer portion using a diluted hydrofluoric acid solution.

[0051] In a ninth aspect, alone or in combination with one or more of the first through eighth aspects, forming the cavity includes forming the cavity using a punch operation, wherein the punch operation is a wet etch operation that forms an isotropic morphology (e.g., the isotropic morphology **250**) in the active area at a bottom surface of the cavity.

[0052] Although FIG. **3** shows example blocks of the method **300**, in some implementations, the method **300** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **3**. In some implementations, the method **300** may include forming the cell sensing structure **200**, an integrated assembly that includes the cell sensing structure **200**, any part described herein of the cell sensing structure **200**, and/or any part described herein of an integrated assembly that includes the cell sensing structure **200**. For example, the method **300** may include forming one or more of the parts the memory cell **100**, including the transistor **105**, the digit line **120**, and/or the cell contact **155**.

[0053] FIGS. **4**A through **4**H are diagrammatic views showing formation of the cell sensing structure **200** at example process stages of an example process **400** of forming the cell sensing structure **200**. In some implementations, the process **400** described below in connection with FIGS. **4**A through **4**H may correspond to the method **300** and/or one or more blocks of the method **300**. However, the process described below is an example, and other example processes may be used to form the cell sensing structure **200**, an integrated assembly that includes the cell sensing structure **200**, and/or one or more parts of the cell sensing structure **200** and/or an integrated assembly including the cell sensing structure **200**.

[0054] As illustrated in FIG. **4**A, the process **400** includes forming the device region **240** with the multi-layer structures **210-1** and **210-2** that extend from the device region **240**. Forming the device region **240** may include forming the semiconductor layer **245**, the dielectric layer **220-8**, and/or the dielectric layer **220-9** using a combination of deposition, photolithography, and etching operations.

[0055] The multi-layer structures **210-1** and **210-2** include the dielectric layers **220-1** through **220-7**. Forming the multi-layer structures **210-1** and **210-2** may include forming the dielectric layers **220-1** through **220-7** using a combination of deposition, photolithography, and etching operations. In some implementations, forming the dielectric layers **220-1** through **220-7** includes forming unadulterated dielectric layers. In some implementations, forming the multi-layer structures **210-1** and **210-2** includes forming an outermost layer (e.g., the dielectric layer **220-7**) using a deposition operation to form a silicon oxycarbide layer. In some implementations, forming the dielectric layer **220-7** includes forming the dielectric layer **220-7** as part of a cap portion **405** in the planarized tip

region **230** (e.g., a portion that fully traverses the digit-line structure **210-1**).
[0056] As further illustrated in FIG. **4**A, forming the multi-layer structures **210-1** and **210-2** includes forming a footer portion **410** that extends away from the base region **225**. In some implementations, and as shown in FIG. **4**A, forming the footer portion **410** includes forming a top segment (e.g., a portion of the layer **220-7**) that is over and/or on a bottom segment (e.g., a portion of the layer **220-6**).
[0057] As illustrated in FIG. **4**B, the process **400** includes forming a temporary, non-conformal layer **415** over the cap portion **405**. In some implementations, forming the temporary, non-conformal layer **415** includes forming the temporary, non-conformal layer **415** using a deposition operation to deposit an oxide material over and/or on the cap portion **405**. In some implementations, operations to form the temporary, non-conformal layer **415** are performed in a chamber of a dry etch tool.
[0058] In some implementations, forming the temporary, non-conformal layer **415** includes forming the temporary, non-conformal layer **415** to have a thickness (D**1**) that is included in a range of approximately 4 nanometers (nm) to approximately 5 nm. If the thickness D**1** is less than approximately 4 nm, the temporary, non-conformal layer **415** may be unable to provide sufficient protection to the cap portion **405** (e.g., a portion of the dielectric layer **220-7** within the cap portion **405**) during a subsequent operation that is intended to damage the footer portion **410** (e.g., a portion of the dielectric layer **220-7** within the footer portion **410**). In such a case, the cap portion **405** may not be preserved during downstream manufacturing and further damage to the cell sensing structure **200** may occur. If the thickness D**1** is between approximately 4 nm and 5 nm, the temporary, non-conformal layer **415** may provide sufficient protection to the cap portion **405**. Additionally, or alternatively and if the thickness D**1** is between approximately 4 nm and 5 nm, the footer portion **410** may remain exposed for the subsequent operation intended to damage the footer portion **410**. Additionally, or alternatively and if the thickness D**1** is greater than approximately 5 nm, the temporary, non-conformal layer **415** may have an increased overhang and/or bridge between adjacent multi-layer structures (e.g., a bridge between the multi-layer structures **210-1** and **210-2**). Such an overhang and/or bridge may to protect the footer portion **410** from the subsequent operation intended to damage the footer portion **410**. However, other values and ranges for the thickness D**1** are within the scope of the present disclosure.
[0059] As illustrated in FIG. **4**C, the process **400** includes damaging the footer portion **410**. Damaging the footer portion **410** may include, for example, using an oxygen-based, directional bias operation to diffuse oxygen into a top segment **420** of the footer portion (e.g., a portion of the dielectric layer **220-7** within the footer portion **410**). The oxygen-based, directional bias operation may include introducing oxygen into a plasma environment and implanting oxygen ions into the top segment **420**. Implanting the oxygen ions in the top segment **420** may reduce a resistance of the top segment **420** to subsequent removal using a wet etch operation. In some implementations, operations to damage the footer portion **410** are performed in the chamber of the dry etch tool used to form the temporary, non-conformal layer **415**.
[0060] As further illustrated in FIG. **4**C, the temporary, non-conformal layer **415** prevents similar damage to the cap portion **405** (e.g., prevents similar damage to the dielectric layer **220-7** within the cap portion **405**). By preventing the damage, a robustness of the cap portion **405** is maintained to prevent removal of the cap portion **405** during the subsequent removal of the top segment **420**.
[0061] As illustrated in FIG. **4**D, the process **400** includes removing the temporary, non-conformal layer **415** and the top segment **420**. Techniques and/or operations to remove the temporary, non-conformal layer **415** and/or the top segment **420** may include using a hydrofluoric acid solution, among other examples. In such a case, the cap portion **405** is resistant to removal and preserved.
[0062] As illustrated in FIG. **4**E, the process **400** includes removing remaining portions of the footer portion **410** (e.g., a bottom segment of the footer portion **410**). In some implementations, techniques and/or operations to remove the remaining portions of the footer portion **410** include

using a diluted hydro fluoric acid solution. Alternatively, and in some implementations, techniques and/or operations to remove the remaining portions of the footer portion **410** include using a hot phosphorous solution. In such cases, the cap portion **405** is resistant to removal and preserved.

[0063] As illustrated in FIG. **4**F, the process **400** includes forming a cavity **425** that penetrates into the semiconductor layer **245** (e.g., into an active area of the device region **240**). In some implementations, techniques to form the cavity **425** include using a punch operation that uses a liquid etchant (e.g., in other words, the punch operation may be a wet etch operation). As described in greater detail in connection with FIG. **5**, the wet etch operation using the liquid etchant may form the isotropic morphology **250** at a bottom surface of the cavity **425**.

[0064] As illustrated in FIG. **4**G, the contact structure **205** is formed between the multi-layer structures **210-1** and **210-2**. Techniques used to form the contact structure **205** may include using deposition operations to deposit the conductive layers **215-1** and **215-2**.

[0065] As illustrated in FIG. **4**H, the cell sensing structure **200** is planarized. Techniques to planarize the cell sensing structure **200** may include using a chemical mechanical planarization (CMP) operation to remove portions of and/or planarize tips of the multi-layer structures **210-1** and **210-2** (e.g., and remove the cap portion(s) **405**). Additionally, or alternatively and in some implementations, the CMP operation may remove portions of and/or planarize the conductive layer **215-1**.

[0066] As indicated above, the process **400** described in connection with FIGS. **4**A through **4**H are provided as an example. Other examples may differ from what is described with respect to FIGS. **4**A through **4**H. The structure shown in FIG. **4**H may be equivalent to the cell sensing structure **200** described elsewhere herein. In process the process steps above that describe forming material, such material may be formed, for example, using chemical vapor deposition, atomic layer deposition, physical vapor deposition, or another deposition operation. In process steps above that describe removing material, such material may be removed, for example, using a wet etching operation (e.g., wet chemical etching), a dry etching operation (e.g., plasma etching), an ion etching operation (e.g., sputtering or reactive ion etching), atomic layer etching, or another etching operation.

[0067] As described in connection with FIGS. **3** and **4**A through **4**H, and in some implementations, a series of semiconductor manufacturing operations includes forming, above a device region (e.g., the device region **240**) of a memory device (e.g., the memory cell **100**), a digit line structure (e.g., the multi-layer structure **210-1**) including a multi-layer structure having a footer portion (e.g., the footer portion **410**) extending away from a base region (e.g., the base region **225**) of the digit line structure and a cap portion (e.g., the cap portion **405**) in a tip region (e.g., the planarized tip region **230**) that extends away from the base region. The series of semiconductor manufacturing operations includes forming a temporary, non-conformal layer (e.g., the temporary, non-conformal layer **415**) over the cap portion. The series of semiconductor manufacturing operation includes damaging a top segment of the footer portion (e.g., the top segment **420**), removing the temporary, non-conformal layer; and removing the footer portion. The series of semiconductor operations includes forming a cavity (e.g., the cavity **425**) that extends into an active area (e.g., the semiconductor layer **245**) of the device region.

[0068] In this way, the cap portion is preserved to increase a process margin associated with subsequent formation of one or more additional features of the memory device, thereby reducing a likelihood of a conductive layer (e.g., the conductive layer **215-3**) being exposed through removal of materials included in one or more layers of the digit line structure. Furthermore, and by reducing the likelihood of the conductive layer being exposed, a potential of electrical shorting within the memory device is reduced, to improve a quality and/or a reliability of the memory device, thereby reducing an amount of resources (e.g., raw materials, labor, semiconductor processing tools, and/or computing resources) used to support a market consuming the memory device.

[0069] FIG. **5** is a diagrammatic view of example etch operations **500** described herein. The etch operations **500** include an isotropic wet etch operation **505** and an anisotropic dry etch operation

510.

[0070] As part of the isotropic wet etch operation **505**, an etchant **515** may remove from a layer **520** material uniformly in all directions. The isotropic wet etch operation **505** may form the isotropic morphology **250** described in connection with FIGS. **2-4**H and elsewhere herein. The isotropic morphology **250** may include surfaces and/or sidewalls that have an approximately spherical, cylindrical, and/or curved shape.

[0071] In contrast, and as part of the anisotropic dry etch operation **510**, an etchant **525** may remove from the layer **520** material in a single direction (e.g., the anisotropic dry etch operation does not remove material in all directions). The anisotropic dry etch operation **510** may form an anisotropic morphology **530**. The anisotropic morphology **530** may include surfaces and/or sidewalls that have approximately planar and/or angled shapes.

[0072] As indicated above, FIG. **5** is provided as an example. Other examples may differ from what is described with respect to FIG. **5**.

[0073] FIG. **6** is a diagrammatic view of an example memory device **600** described herein. The memory device **600** may include a memory array **602** that includes multiple memory cells **604**. A memory cell **604** is programmable or configurable into a data state of multiple data states (e.g., two or more data states). For example, a memory cell **604** may be set to a particular data state at a particular time, and the memory cell **604** may be set to another data state at another time. A data state may correspond to a value stored by the memory cell **604**. The value may be a binary value, such as a binary 0 or a binary 1, or may be a fractional value, such as 0.5, 1.5, or the like. A memory cell **604** may include a capacitor to store a charge representative of the data state. For example, a charged and an uncharged capacitor may represent a first data state and a second data state, respectively. As another example, a first level of charge (e.g., fully charged) may represent a first data state, a second level of charge (e.g., fully discharged) may represent a second data state, a third level of charge (e.g., partially charged) may represent a third data state, and so on.

[0074] Operations such as reading and writing (i.e., cycling) may be performed on memory cells **604** by activating or selecting the appropriate access line **606** (shown as access lines AL **1** through AL M) and digit line **608** (shown as digit lines DL **1** through DL N). An access line **606** may also be referred to as a “row line” or a “word line,” and a digit line **608** may also be referred to a “column line” or a “bit line.” Activating or selecting an access line **606** or a digit line **608** may include applying a voltage to the respective line. An access line **606** and/or a digit line **608** may comprise, consist of, or consist essentially of a conductive material, such as a metal (e.g., copper, aluminum, gold, titanium, or tungsten) and/or a metal alloy, among other examples. In FIG. **6**, each row of memory cells **604** is connected to a single access line **606**, and each column of memory cells **604** is connected to a single digit line **608**. By activating one access line **606** and one digit line **608** (e.g., applying a voltage to the access line **606** and digit line **608**), a single memory cell **604** may be accessed at (e.g., is accessible via) the intersection of the access line **606** and the digit line **608**. The intersection of the access line **606** and the digit line **608** may be called an “address” of a memory cell **604**.

[0075] In some implementations, the logic storing device of a memory cell **604**, such as a capacitor, may be electrically isolated from a corresponding digit line **608** by a selection component, such as a transistor. The access line **606** may be connected to and may control the selection component. For example, the selection component may be a transistor, and the access line **606** may be connected to the gate of the transistor. Activating the access line **606** results in an electrical connection or closed circuit between the capacitor of a memory cell **604** and a corresponding digit line **608**. The digit line **608** may then be accessed (e.g., is accessible) to either read from or write to the memory cell **604**.

[0076] A row decoder **610** and a column decoder **612** may control access to memory cells **604**. For example, the row decoder **610** may receive a row address from a memory controller **614** and may activate the appropriate access line **606** based on the received row address. Similarly, the column decoder **612** may receive a column address from the memory controller **614** and may activate the appropriate digit line **608** based on the column address.

[0077] Upon accessing a memory cell **604**, the memory cell **604** may be read (e.g., sensed) by a sense component **616** to determine the stored data state of the memory cell **604**. For example, after accessing the memory cell **604**, the capacitor of the memory cell **604** may discharge onto its corresponding digit line **608**. Discharging the capacitor may be based on biasing, or applying a voltage, to the capacitor. The discharging may induce a change in the voltage of the digit line **608**, which the sense component **616** may compare to a reference voltage (not shown) to determine the stored data state of the memory cell **604**. For example, if the digit line **608** has a higher voltage than the reference voltage, then the sense component **616** may determine that the stored data state of the memory cell **604** corresponds to a first value, such as a binary 1. Conversely, if the digit line **608** has a lower voltage than the reference voltage, then the sense component **616** may determine that the stored data state of the memory cell **604** corresponds to a second value, such as a binary 0. The detected data state of the memory cell **604** may then be output (e.g., via the column decoder **612**) to an output component **618** (e.g., a data buffer). A memory cell **604** may be written (e.g., set) by activating the appropriate access line **606** and digit line **608**. The column decoder **612** may receive data, such as input from input component **620**, to be written to one or more memory cells **604**. A memory cell **604** may be written by applying a voltage across the capacitor of the memory cell **604**.

[0078] The memory controller **614** may control the operation (e.g., read, write, re-write, refresh, and/or recovery) of the memory cells **604** via the row decoder **610**, the column decoder **612**, and/or the sense component **616**. The memory controller **614** may generate row address signals and column address signals to activate the desired access line **606** and digit line **608**. The memory controller **614** may also generate and control various voltages used during the operation of the memory array **602**.

[0079] In some implementations, the memory device **600** includes the cell sensing structure **200** and/or an integrated assembly that includes the cell sensing structure **200**. For example, the memory array **602** may include the cell sensing structure **200** and/or an integrated assembly that includes the cell sensing structure **200**. Additionally, or alternatively, the memory cell **604** may include a memory cell described elsewhere herein.

[0080] As indicated above, FIG. **6** is provided as an example. Other examples may differ from what is described with respect to FIG. **6**.

[0081] In some implementations, a structure includes a semiconductor layer; a dielectric layer that is proximate to the semiconductor layer; a multi-layer structure that extends away from an approximately planar surface that is across the semiconductor layer and the dielectric layer, comprising; a planarized tip region; and an outer silicon oxycarbide layer that extends from the approximately planar surface to the planarized tip region, wherein a portion of the outer silicon oxycarbide layer in the planarized tip region is unadulterated; and a conductive structure proximate to the multi-layer structure, comprising: a portion that extends through the approximately planar surface and that has a tip with an isotropic morphology, wherein the isotropic morphology includes a curvature traversing a portion of the semiconductor layer and a portion of the dielectric layer.

[0082] In some implementations, an apparatus includes a device region; a digit line structure that extends away from the device region, comprising: a base region; a planarized tip region that is away from the base region; and a multi-layer structure, comprising: a first dielectric layer that extends from the base region to the planarized tip region; a second dielectric layer that conforms to the first dielectric layer and that extends from the bas region to the planarized tip region; a third dielectric layer that conforms to the second dielectric layer and that extends from the base region to the planarized tip region, wherein portions of the first dielectric layer, the second dielectric layer, and the third dielectric layer that are in the planarized tip region are unadulterated; and a cell contact structure that is adjacent to the digit line structure, comprising: a tip that has an isotropic morphology and that extends beyond the base region and into an insulative area and an active area within the device region, wherein the isotropic morphology includes a curvature traversing a portion of the insulative area and a portion of the active area.

[0083] In some implementations, a method includes forming, above a device region of a memory device, a digit line structure including a multi-layer structure having a footer portion extending away from a base region of the digit line structure and a cap portion in a tip region away from the base region; forming a temporary, non-conformal layer over the cap portion; damaging a top segment of the footer portion; removing the temporary, non-conformal layer; removing the footer portion; and forming a cavity that extends into an active area of the device region.

[0084] The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the implementations to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the implementations described herein.

[0085] The orientations of the various elements in the figures are shown as examples, and the illustrated examples may be rotated relative to the depicted orientations. The descriptions provided herein, and the claims that follow, pertain to any structures that have the described relationships between various features, regardless of whether the structures are in the particular orientation of the drawings, or are rotated relative to such orientation. Similarly, spatially relative terms, such as "below," "beneath," "lower," "above," "upper," "middle," "left," and "right," are used herein for ease of description to describe one element's relationship to one or more other elements as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the element, structure, and/or assembly in use or operation in addition to the orientations depicted in the figures. A structure and/or assembly may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein may be interpreted accordingly. Furthermore, the cross-sectional views in the figures only show features within the planes of the cross-sections, and do not show materials behind the planes of the cross-sections, unless indicated otherwise, in order to simplify the drawings.

[0086] As used herein, the terms "substantially" and "approximately" mean "within reasonable tolerances of manufacturing and measurement." As used herein, "satisfying a threshold" may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like. All ranges described herein are inclusive of numbers at the ends of those ranges, unless specifically indicated otherwise.

[0087] Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of implementations described herein. Many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. For example, the disclosure includes each dependent claim in a claim set in combination with every other individual claim in that claim set and every combination of multiple claims in that claim set. As used herein, a phrase referring to "at least one of" a list of items refers to any combination of those items, including single members. As an example, "at least one of: a, b, or c" is intended to cover a, b, c, a+b, a+c, b+c, and a+b+c, as well as any combination with multiples of the same element (e.g., a+a, a+a+a, a+a+b, a+a+c, a+b+b, a+c+c, b+b, b+b+b, b+b+c, c+c, and c+c+c, or any other ordering of a, b, and c).

[0088] No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles "a" and "an" are intended to include one or more items and may be used interchangeably with "one or more." Further, as used herein, the article "the" is intended to include one or more items referenced in connection with the article "the" and may be used interchangeably with "the one or more." Where only one item is intended, the phrase "only one," "single," or similar language is used. Also, as used herein, the terms "has," "have," "having," or the like are intended to be open-ended terms that do not limit an element that they modify (e.g., an element “having” A may also have B). Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. As used herein, the term “multiple” can be replaced with “a plurality of” and vice versa. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

Claims

1. A structure, comprising: a semiconductor layer; a dielectric layer that is proximate to the semiconductor layer; a multi-layer structure that extends away from an approximately planar surface that is across the semiconductor layer and the dielectric layer, comprising; a planarized tip region; and an outer silicon oxycarbide layer that extends from the approximately planar surface to the planarized tip region, wherein a portion of the outer silicon oxycarbide layer in the planarized tip region is unadulterated; and a conductive structure proximate to the multi-layer structure, comprising: a portion that extends through the approximately planar surface and that has a tip with an isotropic morphology, wherein the isotropic morphology includes a curvature traversing a portion of the semiconductor layer and a portion of the dielectric layer.
2. The structure of claim 1, wherein the conductive structure is approximately parallel to the outer silicon oxycarbide layer.
3. The structure of claim 1, wherein the outer silicon oxycarbide layer extends from the approximately planar surface to the planarized tip region.
4. The structure of claim 1, wherein the outer silicon oxycarbide layer is a portion of a multi-layer sidewall structure that is between a conductive layer of the multi-layer structure and the conductive structure.
5. The structure of claim 1, wherein an inflection point near the tip at which the conductive structure transitions to the isotropic morphology aligns with an outer surface of the outer silicon oxycarbide layer.
6. An apparatus, comprising: a device region; a digit line structure that extends away from the device region, comprising: a base region; a planarized tip region that is away from the base region; and a multi-layer structure, comprising: a first dielectric layer that extends from the base region to the planarized tip region; a second dielectric layer that conforms to the first dielectric layer and that extends from the base region to the planarized tip region; a third dielectric layer that conforms to the second dielectric layer and that extends from the base region to the planarized tip region, wherein portions of the first dielectric layer, the second dielectric layer, and the third dielectric layer that are in the planarized tip region are unadulterated; and a cell contact structure that is proximate to the digit line structure, comprising: a tip that has an isotropic morphology and that extends beyond the base region and into an insulative area and an active area within the device region, wherein the isotropic morphology includes a curvature traversing a portion of the insulative area and a portion of the active area.
7. The apparatus of claim 6, wherein the first dielectric layer, the second dielectric layer, and the third dielectric layer include portions that form a sidewall structure adjacent to a digit line of the digit line structure.
8. The apparatus of claim 6, wherein the active area corresponds to a source/drain region of a transistor device.
9. The apparatus of claim 6, wherein the cell contact structure connects with a capacitor of a dynamic random access memory device.
10. The apparatus of claim 6, wherein the digit line structure is a first digit line structure, and further comprising: a second digit line structure adjacent to the first digit line structure, and wherein the cell contact structure is between the second digit line structure and the first digit line

structure.

11. A method, comprising: forming, above a device region of a memory device, a digit line structure including a multi-layer structure having a footer portion that extends away from a base region of the digit line structure and a cap portion that is away from the base region; forming a temporary, non-conformal layer over the cap portion; damaging a top segment of the footer portion; removing the temporary, non-conformal layer; removing the footer portion; and forming a cavity that extends into an insulative area and an active area of the device region.

12. The method of claim 11, wherein forming the multi-layer structure includes: forming an outermost layer of the multi-layer structure using a deposition operation to deposit a layer of silicon oxycarbide over one or more dielectric layers included in the footer portion and the cap portion.

13. The method of claim 11, wherein forming the temporary, non-conformal layer includes: forming the temporary, non-conformal layer using a deposition operation to deposit an oxide material over the cap portion.

14. The method of claim 11, wherein forming the temporary, non-conformal layer includes: forming the temporary, non-conformal layer to have a thickness that is included in a range of approximately 4 nanometers to approximately 5 nanometers above an apex of the cap portion.

15. The method of claim 11, wherein damaging the top segment of the footer portion includes: damaging the top segment of the footer portion using an oxygen-based, directional bias operation.

16. The method of claim 11, wherein removing the temporary, non-conformal layer includes: removing the temporary, non-conformal layer using a hydrofluoric acid solution.

17. The method of claim 11, wherein removing the footer portion includes: removing a top segment of the footer portion using a hydrofluoric acid solution.

18. The method of claim 11, wherein removing the footer portion includes: removing a bottom segment of the footer portion using a hot phosphorous solution.

19. The method of claim 11, wherein removing the footer portion includes: removing a bottom segment of the footer portion using a diluted hydrofluoric acid solution.

20. The method of claim 11, wherein forming the cavity includes: forming the cavity using a punch operation, wherein the punch operation is a wet etch operation that forms an isotropic morphology at a bottom surface of the cavity.